CƠ SỞ DỮ LIỆU · QUẢN LÝ NHÀ TRỌ

PostgreSQL · model49 / V13 | 49 bảng · 455 cột · 151 khóa ngoại | Đối chiếu database ngày 2026-11-01

01 · Tài khoản và phân quyền 6 bảng

users

id	**bigint** NN
username	varchar(50) NN
password_hash	varchar(255) NN
role	varchar(20) NN
status	varchar(20) NN
created_at	timestamptz NN

tenant_profiles

id	**bigint** NN
user_id	bigint
full_name	varchar(150) NN
identity_number	varchar(30)
phone	varchar(30)
email	varchar(150)
date_of_birth	date
hometown	text
created_at	timestamptz NN

employee_assignments

id	**bigint** NN
user_id	bigint NN
property_id	bigint
room_id	bigint
assigned_at	timestamptz NN
revoked_at	timestamptz

staff_permission_grants

id	**bigint** NN
assignment_id	bigint NN
action_code	varchar(40) NN
amount_limit	numeric(14,2)
expires_at	timestamptz
granted_by	bigint NN
created_at	timestamptz NN

password_reset_tokens

id	**bigint** NN
user_id	bigint NN
token_hash	varchar(64) NN
issued_by	bigint NN
expires_at	timestamptz NN
used_at	timestamptz
attempts	smallint NN
created_at	timestamptz NN

audit_events

id	**bigint** NN
actor_user_id	bigint
actor_type	varchar(10) NN
action	varchar(80) NN
resource_type	varchar(80) NN
resource_id	varchar(100)
before_data	jsonb
after_data	jsonb
reason	text
recorded_at	timestamptz NN

02 · Nhà trọ và tài sản 7 bảng

properties

id	**bigint** NN
property_code	varchar(30) NN
name	varchar(150) NN
address	text NN
status	varchar(20) NN
meter_cutoff_day	smallint NN
invoice_issue_day	smallint NN
collection_timing	varchar(20) NN
created_at	timestamptz NN

rooms

id	**bigint** NN
property_id	bigint NN
floor_number	smallint NN
room_code	varchar(30) NN
max_occupants	smallint NN
area_sqm	numeric(8,2)
description	text NN
internal_note	text NN
opened_on	date NN
retired_on	date

room_price_history

id	**bigint** NN
room_id	bigint NN
effective_from	date NN
amount	numeric(14,2) NN
recorded_by	bigint
created_at	timestamptz NN

assets

id	**bigint** NN
asset_type_id	bigint NN
asset_code	varchar(50) NN
quantity	numeric(12,2) NN
status	varchar(20) NN
note	text

room_operating_periods

id	**bigint** NN
room_id	bigint NN
valid_from	timestamptz NN
valid_to	timestamptz
status	varchar(20) NN
recorded_by	bigint
note	text

asset_types

id	**bigint** NN
code	varchar(40) NN
name	varchar(150) NN
unit_name	varchar(30) NN

asset_placements

id	**bigint** NN
asset_id	bigint NN
room_id	bigint
property_id	bigint
valid_from	timestamptz NN
valid_to	timestamptz
recorded_by	bigint

06 · Sự cố và bảo trì 4 bảng

maintenance_schedules

id	**bigint** NN
asset_id	bigint
room_id	bigint
property_id	bigint
title	varchar(150) NN
interval_days	integer NN
next_due	date NN
is_active	boolean NN
created_by	bigint

maintenance_requests

id	**bigint** NN
room_id	bigint
property_id	bigint
asset_placement_id	bigint
schedule_id	bigint
scheduled_for	date
submitted_by	bigint NN
assigned_to	bigint
title	varchar(150) NN
description	text NN
priority	varchar(20) NN
status	varchar(20) NN
incident_type	varchar(50) NN
expected_date	date NN
created_at	timestamptz NN

maintenance_events

id	**bigint** NN
request_id	bigint NN
action	varchar(30) NN
actor_id	bigint NN
note	text NN
recorded_at	timestamptz NN

maintenance_costs

id	**bigint** NN
request_id	bigint NN
description	text NN
amount	numeric(14,2) NN
incurred_on	date NN
advanced_by_tenant_id	bigint
recorded_by	bigint NN
created_at	timestamptz NN

07 · Thông báo 2 bảng

notifications

id	**bigint** NN
notification_type	varchar(40) NN
title	varchar(200) NN
body	text NN
billing_document_id	bigint
contract_id	bigint
maintenance_request_id	bigint
consumption_alert_id	bigint
deduplication_key	varchar(180) NN
created_by	bigint
sent_at	timestamptz NN

notification_recipients

id	**bigint** NN
notification_id	bigint NN
user_id	bigint NN
read_at	timestamptz

08 · Tệp đính kèm 2 bảng

stored_files

id	**bigint** NN
storage_key	varchar(255) NN
original_name	varchar(255) NN
content_type	varchar(100) NN
size_bytes	bigint
sha256	varchar(64)
external_url	text
uploaded_by	bigint NN
uploaded_at	timestamptz NN

file_attachments

id	**bigint** NN
file_id	bigint NN
room_id	bigint
asset_id	bigint
handover_item_id	bigint
meter_reading_id	bigint
maintenance_request_id	bigint
maintenance_cost_id	bigint
contract_term_id	bigint
billing_document_id	bigint
violation_id	bigint
fund_entry_id	bigint
caption	text NN
sort_order	integer NN
attached_by	bigint NN
created_at	timestamptz NN

03 · Hợp đồng và cư trú 12 bảng

reservations

id	**bigint** NN
room_id	bigint NN
tenant_id	bigint
reservation_code	varchar(60) NN
contact_name	varchar(150)
contact_phone	varchar(30)
start_date	date NN
end_date	date NN
expires_at	timestamptz NN
status	varchar(20) NN
conditions	text NN
created_by	bigint
created_at	timestamptz NN

contracts

id	**bigint** NN
room_id	bigint NN
reservation_id	bigint
previous_contract_id	bigint
continuation_reason	varchar(30)
contract_code	varchar(60) NN
start_date	date NN
reconciliation_status	varchar(20) NN
status	varchar(20) NN
created_by	bigint
created_at	timestamptz NN

contract_parties

id	**bigint** NN
contract_id	bigint NN
tenant_id	bigint NN
party_role	varchar(30) NN
effective_from	date NN
effective_to	date
agreement_note	text NN

contract_contacts

id	**bigint** NN
contract_id	bigint NN
tenant_id	bigint NN
contact_role	varchar(20) NN
effective_from	date NN
effective_to	date
can_view_billing	boolean NN
authorized_by	bigint

contract_residencies

id	**bigint** NN
contract_id	bigint NN
tenant_id	bigint NN
effective_from	date NN
effective_to	date
relationship	varchar(50)
recorded_by	bigint

contract_terms

id	**bigint** NN
contract_id	bigint NN
version_no	integer NN
effective_from	date NN
lease_end	date NN
rental_price	numeric(14,2) NN
deposit_required	numeric(14,2) NN
payment_day	smallint NN
collection_timing	varchar(20) NN
terms	text NN
signed_at	timestamptz
approved_by	bigint
origin	varchar(20) NN
status	varchar(20) NN
created_at	timestamptz NN

move_out_notices

id	**bigint** NN
contract_id	bigint NN
supersedes_notice_id	bigint
requested_date	date NN
reason	text NN
status	varchar(20) NN
submitted_by	bigint NN
created_at	timestamptz NN

room_possessions

id	**bigint** NN
contract_id	bigint NN
taken_at	timestamptz NN
returned_at	timestamptz
recorded_by	bigint
note	text

handovers

id	**bigint** NN
contract_id	bigint NN
possession_id	bigint
handover_code	varchar(60) NN
handover_type	varchar(20) NN
occurred_at	timestamptz NN
status	varchar(20) NN
given_by	bigint
received_by_tenant_id	bigint
confirmed_by	bigint
confirmed_at	timestamptz
note	text

handover_items

id	**bigint** NN
handover_id	bigint NN
asset_placement_id	bigint
line_no	integer NN
item_name_snapshot	varchar(150) NN
quantity	numeric(12,2) NN
condition_note	text NN
note	text

contract_violations

id	**bigint** NN
contract_id	bigint NN
violation_type	varchar(50) NN
description	text NN
occurred_on	date NN
status	varchar(20) NN
resolution_note	text
created_by	bigint NN
created_at	timestamptz NN

checkout_records

id	**bigint** NN
contract_id	bigint NN
return_handover_id	bigint
billing_end	date NN
status	varchar(20) NN
deposit_snapshot	numeric(14,2) NN
advance_snapshot	numeric(14,2) NN
outstanding_snapshot	numeric(14,2) NN
approved_by	bigint
approved_at	timestamptz
notes	text

04 · Dịch vụ và điện nước 9 bảng

service_items

id	**bigint** NN
service_code	varchar(30) NN
name	varchar(100) NN
unit_name	varchar(30) NN
calculation_type	varchar(20) NN
is_mandatory	boolean NN
is_active	boolean NN

service_price_configs

id	**bigint** NN
property_id	bigint NN
service_item_id	bigint NN
effective_from	date NN
unit_price	numeric(14,2) NN
recorded_by	bigint
created_at	timestamptz NN

contract_service_subscriptions

id	**bigint** NN
contract_id	bigint NN
service_item_id	bigint NN
effective_from	date NN
effective_to	date
quantity	integer NN
charge_mode	varchar(20) NN
recorded_by	bigint

meters

id	**bigint** NN
room_id	bigint NN
service_item_id	bigint NN
meter_code	varchar(60) NN
installed_at	timestamptz NN
retired_at	timestamptz
recorded_by	bigint

meter_readings

id	**bigint** NN
meter_id	bigint NN
observed_at	timestamptz NN
reading_value	numeric(14,3) NN
time_precision	varchar(30) NN
revision_no	integer NN
supersedes_reading_id	bigint
handover_id	bigint
recorded_by	bigint NN
reason	text
recorded_at	timestamptz NN
correction_item_id	bigint

consumption_segments

id	**bigint** NN
start_reading_id	bigint NN
end_reading_id	bigint NN
contract_id	bigint
responsibility	varchar(10) NN
supersedes_segment_id	bigint
recorded_by	bigint
created_at	timestamptz NN

alert_policies

id	**bigint** NN
service_item_id	bigint NN
property_id	bigint
room_id	bigint
effective_from	date NN
history_months	smallint NN
minimum_days	integer NN
threshold_percentage	numeric(9,2) NN
absolute_daily_threshold	numeric(14,3)

consumption_alerts

id	**bigint** NN
segment_id	bigint NN
policy_id	bigint NN
supersedes_alert_id	bigint
baseline_daily_average	numeric(18,6)
applied_threshold	numeric(9,2) NN
method	varchar(40) NN
status	varchar(30) NN
context_note	text
checked_by	bigint
checked_at	timestamptz
created_at	timestamptz NN

alert_baseline_sources

id	**bigint** NN
alert_id	bigint NN
source_segment_id	bigint NN

05 · Hóa đơn và dòng tiền 7 bảng

billing_documents

id	**bigint** NN
contract_id	bigint
reservation_id	bigint
checkout_id	bigint
related_document_id	bigint
document_code	varchar(80) NN
document_type	varchar(30) NN
billing_period	date
issue_date	date NN
due_date	date NN
status	varchar(20) NN
reason	text NN
created_by	bigint
issued_by	bigint
issued_at	timestamptz
created_at	timestamptz NN

billing_items

id	**bigint** NN
document_id	bigint NN
line_no	integer NN
item_kind	varchar(30) NN
direction	varchar(10) NN
adjusts_item_id	bigint
contract_term_id	bigint
subscription_id	bigint
segment_id	bigint
service_price_config_id	bigint
handover_item_id	bigint
maintenance_cost_id	bigint
violation_id	bigint
item_name_snapshot	varchar(150) NN
unit_name_snapshot	varchar(40) NN
unit_price_snapshot	numeric(14,4) NN
calculation_basis	varchar(20) NN
quantity_snapshot	numeric(18,6)
calculation_period	date
service_from	date NN
service_to	date NN
verification_status	varchar(20) NN
basis_note	text NN
source_key	varchar(120)

cash_accounts

id	**bigint** NN
property_id	bigint
account_code	varchar(60) NN
name	varchar(150) NN
account_type	varchar(20) NN
is_active	boolean NN

fund_entries

id	**bigint** NN
entry_code	varchar(120) NN
entry_type	varchar(30) NN
cash_transaction_id	bigint
from_fund_id	bigint
to_fund_id	bigint
debit_item_id	bigint
credit_item_id	bigint
original_entry_id	bigint
amount	numeric(14,2) NN
reason	text NN
posted_by	bigint NN
posted_at	timestamptz NN

cash_transactions

id	**bigint** NN
cash_account_id	bigint NN
payer_or_payee_id	bigint
transaction_code	varchar(80) NN
direction	varchar(5) NN
amount	numeric(14,2) NN
method	varchar(30) NN
occurred_at	timestamptz NN
recorded_at	timestamptz NN
status	varchar(20) NN
external_reference	varchar(150)
counterparty_note	text
reason	text
recorded_by	bigint NN
confirmed_by	bigint
confirmed_at	timestamptz
payment_document_id	bigint

fund_accounts

id	**bigint** NN
fund_code	varchar(100) NN
fund_type	varchar(20) NN
reconciliation_status	varchar(20) NN
contract_id	bigint
reservation_id	bigint
unapplied_receipt_id	bigint
beneficiary_tenant_id	bigint
created_at	timestamptz NN

fund_holds

id	**bigint** NN
fund_account_id	bigint NN
related_billing_item_id	bigint
amount	numeric(14,2) NN
reason	text NN
review_on	date NN
created_by	bigint NN
released_at	timestamptz
created_at	timestamptz NN

Khóa và quan hệ: ký hiệu khóa = PK · nhánh chân quạ = nhiều · vòng tròn = phía quan hệ tùy chọn. Tên bảng, cột và kiểu dữ liệu giữ theo PostgreSQL.
Phạm vi: 49 bảng nghiệp vụ; không đưa flyway_schema_history và 10 SQL view vào thành bảng. 132 CHECK, 7 EXCLUDE và chỉ mục có điều kiện được lưu trong DBML/ghi chú.
Nguồn: pg_catalog của hostel_review49 sau V1–V13; sơ đồ xuất bằng dbdiagram.io. Tiêu đề nhóm và chú giải được đặt bên ngoài bản xuất.